

Attachment 1 Specifications

Controlled Atmosphere Storage System – Kearneysville WV

The USDA, Agricultural Research Service (ARS), Appalachian Fruit Research Station (AFRS), Innovative Fruit Production, Improvement and Protection unit are requesting a controlled atmosphere system that can be retrofitted into our existing cold rooms used to store pome fruits. The controlled atmosphere system is required for a horticultural laboratory setting to conduct research and analysis on postharvest storage of fruit. Its primary function will be to create, maintain, and monitor precisely controlled atmospheric environments of fruit samples for materials testing and other scientific studies. This will allow researchers to study the effects of specific gas concentrations (such as oxygen, carbon dioxide, and nitrogen) on various subject fruits in a contained, observable, and repeatable manner.

Background and Scope

Modern storage of pome fruits (apple and pears) has been revolutionized with the advent of (dynamic) controlled atmospheric storage systems. These systems modify the atmosphere of the storage room or vessel by removing Oxygen and CO₂, while increasing N concentrations. As a result of the modified atmosphere, pome fruits can be stored for months beyond their standard shelf life in cold storage. For pome fruit breeding programs, the evaluation of storage potential is a critical trait in determining the potential of a new variety. However, AFRS lacks controlled atmosphere storage, so we are unable to properly evaluate our breeding material under commercial storage conditions. Thus, it is imperative that we obtain a system to retrofit into our cold storage rooms to properly evaluate breeding material for shelf-life and storage capabilities.

One Each. Brand Name or Equal to Storage Control Systems LabPod CA System

Salient Characteristics

Modular and Independent Control: The system must consist of multiple, independent chambers (or "pods") where the atmospheric conditions in each chamber can be controlled separately from the others. This allows for simultaneous experiments with different variables.

Automated Atmosphere Control: The system must automatically measure and maintain user-defined setpoints for gas concentrations (e.g., O₂, CO₂). It must be a closed-loop system that injects the necessary gases to correct any deviation from the setpoint.

Integrated Sensing and Data Logging: Each chamber must contain integrated sensors for O₂ and CO₂. The system must be capable of logging this data over time and allowing for data export (e.g., to a CSV or Excel file) for analysis.

Sealed Environment: Each chamber must be hermetically sealed to prevent gas leakage and contamination from the ambient environment, ensuring the integrity of the experiment.

Deployed in to existing infrastructure: The controls system and chambers must be compatible for placement in existing climate-controlled facilities (ie. Cold room that operated between 0 – 4 degrees C).

Physical Characteristics Chamber Construction: Chambers must be constructed from a durable, clear material (e.g., acrylic or polycarbonate) to allow for unobstructed visual observation of the contents.

Design: The system should have a compact, stackable design to conserve space.

Access Ports: Each chamber must include sealed ports for the introduction and removal of samples, as well as for gas lines and sensor connections, without compromising the internal

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atmosphere. **Gaskets and Seals:** All doors, lids, and ports must be fitted with high-quality, airtight gaskets to ensure a positive seal.

Performance Characteristics

Gas Control Range: The system must be able to control gas concentrations within the following approximate ranges: Oxygen (O₂): 0.1% to 30% Carbon Dioxide (CO₂): 0% to 30%

Control Accuracy: The system must maintain gas concentration setpoints with an accuracy of at least ±0.2% or better.

Control Interface: The system must be operated via a central controller with an intuitive user interface (e.g., a touchscreen) that allows for programming, monitoring, and control of all connected chambers.

Flow Control: The system must include precision gas flow control to manage the injection of nitrogen, CO₂, and air to maintain the desired atmospheric blend.

Safety Alarms: The system must feature audible and visual alarms to notify users of deviations from setpoints, gas supply issues, or other system errors.